

Operational Blueprinting & Application Build Extension

Dynamic Operational Synchronization & Scheduling Platform

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PURPOSE This extension translates the validated operational blueprint into a client-facing application build plan, including Vector proprietary scheduling and scenario scaffolding, phased delivery blocks, operating-window logic, integration assumptions and governed deployment boundaries.

Contents

1. Executive Summary
2. Business Use Case
3. Core Scheduling Philosophy
4. Vector Proprietary Scheduling And Scenario Scaffolding
5. Product Development Layers
6. Phase Of Development In Sprint Level
7. Working Flow Details
8. Application Modules
9. Assumptions Made To Determine The Flow
10. Proposed Implementation Architecture
11. External Data Sources Assumed To Be Available
12. Build Deliverables
13. Implementation Clarification
14. Important Build Boundaries
15. Reference Inputs

1. Executive Summary

The Operational Blueprinting and Simulation Prototype engagement establishes the operating logic, data assumptions, decision workflows and implementation priorities for the ABL - Berau Coal transshipment environment. This extension defines how that validated blueprint moves into the application build.

The proposed product is a **Dynamic Operational Synchronization & Scheduling Platform**. It will convert Berau shipment demand into an executable ABL logistics plan by coordinating cargo readiness, jetty loading, tug-barge movement, tide and bridge windows, CTS availability, OGV loading sequence, exception recovery, approval and publication.

The platform is not only a fleet-tracking screen, map view or digital Excel replacement. It is a governed scheduling and simulation application designed to place each shipment movement through eligible route steps, available operating windows and approved decision workflows.

Vector will use its proprietary scheduling, scenario and dynamic-constraint scaffolding as the base for the application build. The ABL-specific implementation will configure and extend this base around Berau demand, ABL assets, tide and bridge logic, port/CTS operations, recovery workflows, approval governance and required integrations.

2. Business Use Case

The business use case is:

OPERATING PRINCIPLE Synchronize Berau Coal's shipment commitments with ABL's real operational flow capacity so that cargo readiness, asset availability, tide/bridge windows, jetty/CTS capacity, OGV loading sequence and recovery decisions are planned and governed from one shared operating platform.

2.1 Business problems addressed

Business problem	Application response
Static Excel planning cannot absorb daily and intra-day changes	Structured demand intake, versioned plan generation, conflict checks and governed replanning
Shipment demand is not continuously synchronized with logistics capacity	OGV demand, cargo grade, route, tug-barge, jetty, CTS and operating-window capacity are held in one planning model

Business problem	Application response
Asset position is visible but operational status is unclear	GPS/AIS evidence is separated from confirmed operational events and schedule authority
Tide, bridge and berth windows keep changing the feasible plan	Movement windows are treated as schedulable capacity
Coal grade and loading sequence affect execution	Cargo requirement and layer sequence are first-class planning inputs
Existing work in progress affects new planning	Active plans, sailing barges, loaded barges, queues and confirmed actuals are plotted before new planning decisions
Disruption recovery is manual and experience-driven	Scenario simulation and recovery recommendations show options, impact and approval path
Approvals and operating decisions require traceability	Dual-party approvals, immutable published snapshots, audit events and governed exports

3. Core Scheduling Philosophy

The scheduling logic is based on a flow principle:

OPERATING PRINCIPLE Each shipment movement is treated as a schedulable entity that must pass through a defined route of operational steps. Each step needs the right resource, enough available capacity and a valid operating window.

For ABL, the route is operational as well as physical:

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Cargo readiness
-> Jetty / BLC loading
-> Tug-barge departure
-> Tide / river movement
-> Bridge crossing
-> CTS / anchorage queue
-> CTS discharge
-> OGV hatch / layer completion
-> Survey / handover closure

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The controlling constraint shifts during the day between tide, bridge, jetty, CTS, berth, tug-barge availability, cargo readiness and OGV laycan pressure. The application will therefore recalculate feasibility around moving operating windows and current execution state, not only around static planned dates.

4. Vector Proprietary Scheduling And Scenario Scaffolding

Vector owns a proprietary base scaffolding for constraint-aware scheduling, scenario simulation and dynamic-capacity planning. This base will be used as the starting point for the ABL application build.

4.1 Capabilities provided by the base scaffolding

Capability	Role in ABL build
Scheduling entity model	Represents shipment movements, route steps, dependencies, priorities and target dates
Route and resource eligibility	Matches each movement with allowed jetties, routes, tug-barge pairs, CTS assets and operating steps
Capacity-window placement	Places movements into free operating windows and prevents invalid overlap
Current-state loading	Considers active work, occupied resources, queues and confirmed actuals before placing new work
Dynamic constraint evaluation	Rechecks feasibility when tide, bridge, berth, asset or cargo readiness changes
Scenario engine	Simulates delay, outage, window-change, reassignment and recovery assumptions
Candidate comparison	Evaluates alternate paths and recovery options against feasibility and impact
Exception and blocker output	Identifies the first operating blocker and the affected downstream commitments

4.2 ABL-specific configuration and extension

The ABL build will configure this base around:

- Berau OGV demand, cargo quantity, coal grade and layer sequence
- ABL jetties, BLC loading capability, tug-barge fleet and CTS assets
- tide windows, bridge windows, river movement and route cycle times
- cargo readiness, stockpile/source eligibility and quality-release status
- active trip state, loaded barges, queues and confirmed operational events
- exception categories, recovery rules, approvals and publishability gates
- reporting, dashboard, audit and export requirements.

The proprietary base accelerates implementation while the blueprinting phase ensures that ABL-specific business rules, data interfaces and governance logic are validated before full deployment.

5. Product Development Layers

5.1 Layer 1 - Governed scheduling model

This layer creates the structured planning spine:

- OGV voyage and laycan intake
- cargo quantity, coal grade and loading sequence

- source, stockpile, jetty and route master data
- tug, barge and CTS availability
- tide and bridge calendars
- deterministic schedule generation
- conflict detection
- plan versioning
- approval, publish and audit.

5.2 Layer 2 - Simulation and operational evidence

This layer adds scenario and live-state interpretation:

- delay, outage, window-change and reassignment assumptions
- projected trip, event, OGV completion and utilization impact
- GPS/AIS-style position evidence
- geofence, ETA variance and stale-signal alerts
- event candidates from device, operator or feed input
- trusted confirmation before schedule actualization.

5.3 Layer 3 - Recovery and guided execution

This layer adds guided decision support:

- recovery input snapshots from active plan, alerts, confirmed events and conflicts
- ranked recovery options
- root-cause validation
- scenario creation from selected recovery option
- publishability assessment
- guided operator workflow
- projection-only commercial and telemetry-trust review surfaces.

6. Phase Of Development In Sprint Level

Sprint level	Build focus	Key outcome
Sprint 0	Application foundation: web application, API service, database, background processing, file storage and routing layer	Runnable product spine
Sprint 1	Master data and governance: organizations, roles, audit, locations, cargo, jetties, routes, assets and compatibility	Safe multi-party operating model
Sprint 2	Demand and capacity intake: OGV demand, cargo layer, asset windows, jetty windows, tide windows and bridge windows	Structured inputs for planning
Sprint 3	Scheduling entity and route model: trips, assignments, schedule events and route-step dependencies	Movement chain represented as schedulable work

Sprint level	Build focus	Key outcome
Sprint 4	Capacity placement and feasibility: candidate paths, occupied/free windows, conflict checks and movement candidates	Feasible plan with explicit blockers
Sprint 5	Plan governance: versions, overrides, approval requests, approval decisions, published snapshots and exports	Approved execution contract
Sprint 6	Scenario simulation: assumptions, projected events, projected trips, constraint evaluations and utilization outputs	What-if planning and impact visibility
Sprint 7	Live evidence: telemetry sources, asset identities, position pings, latest asset state, geofences and ETA variance	Planned-vs-observed movement visibility
Sprint 8	Confirmed operations: integration feeds, devices, event candidates, confirmed events, actualization and device health	Trusted execution status layer
Sprint 9	Recovery recommendations: input snapshots, ranked options, recovery actions and proof pack	Decision support for disrupted flow
Sprint 10	Guided recovery and hardening: workflow guidance, root-cause validation, publishability, review surfaces and tests	Pilot-ready governed control tower

The first release will prove the governed scheduling spine and operating-window logic. Later releases will progressively add simulation, live evidence, trusted field events, recovery guidance and management review surfaces.

6.1 Client Delivery Blocks

Delivery block	Release 1 - Governed scheduling spine
Sprint coverage	Sprint 0 to Sprint 5
System screens	Login and role-based navigation, master data, OGV demand, cargo layer sequence, tide/bridge windows, tug-barge assignment, jetty loading, CTS operations, conflict review, approval, published plan, audit and export screens
Capability	Structured demand intake, master-data setup, operating-window entry, route/resource eligibility, candidate movement generation, feasibility checks, conflict output, plan versioning, dual approval, manual publish and governed export
ABL-owned product	A working governed scheduling application that replaces spreadsheet-only planning for the core ABL-Berau flow and creates an approved execution contract

Delivery block	Release 2 - Scenario and impact simulation
Sprint coverage	Sprint 6
System screens	Simulation workspace, scenario assumption entry, baseline-versus-scenario comparison, projected trip/event impact, constraint impact and utilization review screens

Delivery block	Release 2 - Scenario and impact simulation
Capability	What-if planning for delay, outage, window change and reassignment; projected downstream impact; scenario promotion into governed approval path
ABL-owned product	A planning simulation product that allows ABL to test disruption impact before changing the published plan

Delivery block	Release 3 - Live evidence and confirmed operations
Sprint coverage	Sprint 7 to Sprint 8
System screens	Live resource map, asset signal detail, tracking alert list, operations event console, device/feed health, jetty/CTS actualization and exception linkage screens
Capability	GPS/AIS-style evidence ingestion, latest asset state, geofence/ETA variance, stale-signal alerts, event candidate capture, event confirmation, actual execution update and trusted exception creation
ABL-owned product	An execution-monitoring layer that connects the approved schedule with observed and confirmed field reality

Delivery block	Release 4 - Recovery guidance and control tower hardening
Sprint coverage	Sprint 9 to Sprint 10
System screens	Recovery recommendation console, ranked option detail, root-cause validation, publishability review, guided workflow panel, management review surfaces and hardened audit evidence screens
Capability	Ranked recovery options, recovery proof pack, guided next action, publishability validation, telemetry-trust review and projection-only commercial visibility
ABL-owned product	A governed flow-management control tower that supports disruption recovery from exception identification to approved recovery publication

7. Working Flow Details

7.1 Clean planning flow

```
Import OGV demand
-> Review cargo grade and layer sequence
-> Enter operating windows
-> Generate movement candidates
-> Generate draft plan
-> Review conflicts
-> Submit approval
-> Complete Berau / ABL approvals
-> Run publishability check
-> Manually publish
-> Generate governed export
```

7.2 Disruption and recovery flow

```
Open active exception
-> Generate recovery options
-> Review ranked candidates
-> Validate whether the option resolves the operating cause
-> Create scenario from selected option
-> Run simulation
-> Promote feasible scenario
-> Repair remaining blockers if required
-> Submit approval
-> Complete approvals
-> Run publishability check
-> Manually publish recovery plan
```

7.3 Capacity placement logic

At planning time, the platform will:

1. load current demand and master data;
2. plot current work and already occupied capacity;
3. identify eligible resources for each movement;
4. generate candidate paths;
5. search for continuous feasible windows;
6. mark selected windows as occupied;
7. record conflicts where no feasible placement exists;

8. preserve reason codes and audit lineage.

7.4 Live evidence and actualization

GPS/AIS or synthetic position evidence

- > Normalize asset identity
- > Store raw evidence
- > Derive latest state, geofence and ETA variance
- > Raise alert where required
- > Use alert as scenario input when replanning is needed

Device/operator/event candidate

- > Match to planned trip or schedule event
- > Apply confidence, trust and dedupe rules
- > Confirm or reject
- > Update actual event only after confirmation
- > Create exception if variance requires action
- > Record audit trail

8. Application Modules

Module	Working responsibility
Organizations and access control	Multi-party users, roles, permissions, data scope and approval authority
Master data	Locations, coal grades, mines, stockpiles, jetties, tug/barge/CTS assets, routes, loading rates and compatibility
Planning	OGV voyages, cargo requirements, cargo layer sequence, availability windows, tide windows, bridge windows and import jobs
Scheduling	Plan, plan version, trip, assignment, schedule event, conflict, override, approval, publish, export and simulation
Telemetry	Source registry, asset mapping, position evidence, geofence zones, latest state, ETA projection and tracking alerts
Operations	Integration feeds, device endpoints, health snapshots, event candidates, confirmed events and actualization
Guided workflow	Operator guidance, action enablement and flow evidence
Audit	Request and domain audit trail for governed actions
Operator cockpit	Dense operational cockpit covering demand, cargo sequence, assignments, jetty, CTS, tide/bridge, exceptions, simulation, approvals, published plan, map, recovery, audit and admin

9. Assumptions Made To Determine The Flow

1. Berau provides OGV demand, laycan, cargo quantity, coal grade and layer sequence in a consistent file or integration-ready format.
2. ABL and Berau will agree master-data naming for jetties, routes, assets, grades, locations and operating zones.
3. Tide and bridge windows are initially entered manually or uploaded before live sensor integration is available.
4. Resource capacity includes both time availability and operational eligibility.
5. Active work in progress is plotted before new schedules or recovery options are generated.
6. GPS/AIS or Spinergie-style fleet data is evidence and does not directly change schedule authority.
7. Field events such as loading start, loading completion, bridge crossing, tide gate pass and CTS discharge require confirmation before actualization.
8. Published plans are immutable; replanning creates successor versions.
9. Approval requires Berau and ABL authority before publication.
10. Manual override requires reason capture and audit.
11. Recovery recommendations do not bypass simulation, approval or manual publish.
12. External chartered assets require readiness and lead-time assumptions.
13. Commercial exposure is limited to planning projection, while final demurrage, despatch, invoicing and settlement remain separately scoped.
14. Synthetic seed and replay data are valid for blueprint, prototype, UAT and integration-contract proof before production feeds are connected.

10. Proposed Implementation Architecture

Layer	Technology
Operator web application	React, Vite, TypeScript
Web application testing	Vitest, Testing Library, jsdom
API and business services	Django, Django REST Framework
Background jobs	Celery, Celery Beat
Application server	Gunicorn
Service testing	pytest, pytest-django
Database	PostgreSQL with PostGIS
Cache / broker	Redis
Object storage	MinIO / S3-compatible storage
Edge routing	Nginx
Runtime approach	Containerized deployment model

Deferred integration technology includes, where required by confirmed integration scope, MQTT, HTTP callbacks, AIS/NMEA feeds, Kafka/Redpanda, TimescaleDB or ClickHouse, OPC-UA or Modbus, and live UI push updates.

11. External Data Sources Assumed To Be Available

External data source	Assumed content	Initial handling	Later integration direction
Existing Excel schedule	Current OGV plan and operating schedule	Import or manual upload	Governed file ingestion or direct integration
OGV schedule	ETA, ETB, laycan, quantity, priority and vessel status	Manual/imported demand	API feed from Berau or shipping system
Cargo/source data	Mine, CPP, stockpile, grade, product, available quantity, quality release and jetty eligibility	Master data and demand setup	Cargo/quality or ERP integration
Cargo layer sequence	Hatch/layer order, grade requirement and substitution rule	Cargo layer model	Structured demand feed
Jetty/BLC data	Loading rate, queue, working hours, equipment status and readiness	Manual windows/status	Terminal, PLC or operator console
ABL fleet data	Tug, barge, CTS capacity, compatibility, status, location and availability	Master data and availability windows	Spinergie, GPS/AIS or fleet API
Tide data	Tide tables, height, safe movement windows and water-level observations	Manual tide windows	Sensor/API feed
Bridge data	Opening schedule, crossing rule and queue	Manual bridge windows	Bridge operator console or device feed
Weather/marine data	Rain, wind, wave, visibility and closure alerts	Exception input or planning note	Weather API or local station
GPS/AIS data	Position, speed, heading, timestamp, signal quality and external asset identity	Synthetic replay or vendor-shaped payload	Spinergie, AIS receiver, tracker or vendor API
Operational events	Loading start/end, discharge start/end, stoppage, breakdown and crossing confirmation	Synthetic feed and operator confirmation	MQTT, HTTP callback, PLC, weighbridge, tablet or edge gateway
Device health	Last seen, power, battery, network, firmware and signal gap	Device health snapshot	IoT/device-management integration
Survey and documents	Draft survey, sampling, certificates, SOF and final quantity/quality documents	Deferred milestone visibility	Document management or survey integration
Commercial parameters	Laycan risk, exposure proxy and customer-safe ETA projection	Projection-only review	Separate commercial sign-off

12. Build Deliverables

Deliverable	Description
Vector scheduling and scenario base scaffolding	Proprietary scheduling, scenario and dynamic-constraint foundation owned by Vector and adapted for ABL operations
Application foundation	Containerized web application, API service, database, Redis, object storage and routing layer
Master and capacity model	Assets, routes, eligibility, calendars, operating windows and compatibility
ABL scheduling configuration	Shipment movement entity, candidate paths, capacity placement, moving-window checks and conflict output
Scenario and recovery configuration	What-if assumptions, projected impacts, ranked recovery options and proof pack
Operational cockpit	Role-aware UI for demand, cargo sequence, assignments, jetty, CTS, tide/bridge, exceptions, simulation, approvals, published plan, map, recovery, audit and admin
Evidence ingestion layer	GPS/AIS-shaped telemetry and operational event candidate contracts
Trust and actualization layer	Confirmation, rejection, actual event updates and device/feed health

Deliverable	Description
Governance layer	Access control, dual approval, publishability, immutable snapshots and governed exports
Operator guidance layer	Guided workflow and next-action support
Evidence and test pack	Service tests, web application tests, browser evidence, proof commands and runbooks

13. Implementation Clarification

This extension defines the application build direction that follows the blueprinting and prototype scope.

Final implementation timeline, team size and commercials will be confirmed after:

- ABL and Berau validate the operational rules
- required external data sources are confirmed
- integration owners and access methods are known
- master-data ownership is agreed
- pilot users and approval authorities are named
- UAT scope and deployment environment are confirmed
- licensing and usage terms for Vector proprietary scheduling and scenario scaffolding are finalized.

14. Important Build Boundaries

1. GPS/AIS is evidence, not automatic truth.
2. Raw pings do not update the execution plan directly.
3. Confirmed field events update actual execution state.
4. Recovery options are advisory until simulated, approved and checked for publish readiness.
5. Published plans remain immutable.
6. Customer-facing visibility comes after internal governance is stable.
7. Commercial projection is not final settlement logic.
8. Synthetic proof data is acceptable for prototype and UAT, but production claims require real integration testing.
9. Planned, observed, confirmed, projected and recommended states remain separate.

15. Reference Inputs

This extension is based on:

- the Operational Blueprinting & Simulation Prototype Proposal
- the Scheduling Simulation BRD
- ABL and Berau Coal operating context
- observed transshipment planning and fleet-coordination challenges

- cargo, route, asset, tide, bridge, CTS and OGV scheduling logic
- the shared scheduling principle that work must be placed through eligible route steps, available resources and continuous operating windows
- the product-development direction for the ABL flow-management platform.